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SEMICONDUCTOR PACKAGE AND METHOD FOR MANUFACTURING THE SAME

Abstract

A semiconductor package according to an embodiment includes a first semiconductor die, a plurality of bonding pads on the first semiconductor die, a plurality of connection members on the plurality of bonding pads, wherein each connection member of the plurality of connection members is disposed on a corresponding bonding pad of the plurality of bonding pads, a second semiconductor die on the plurality of connection members, and an anisotropic conductive film between the first semiconductor die and the second semiconductor die, and between the plurality of bonding pads and the plurality of connection members, including an insulating film, and conductive particles inside the insulating film, wherein the conductive particles are positioned between each of the plurality of connection members and the corresponding bonding pad of the plurality of bonding pads.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to and the benefit of Korean Patent Application No. 10-2024-0030271 filed in the Korean Intellectual Property Office on Feb. 29, 2024, the entire contents of which are incorporated herein by reference.

BACKGROUND OF THE INVENTION

(a) Field of the Invention

[0002] The present disclosure relates to a semiconductor package and a method for manufacturing the same.

(b) Description of the Related Art

[0003] The semiconductor industry is pursuing down-sizing, lightening, and thinning the semiconductor packages mounted in electronic devices, while simultaneously pursuing higher speed, multifunction, and large capacity, in response to the demand for down-sizing and lightening of electronic devices. Therefore, the need for packaging technology capable of storing more data and transmitting data at a higher speed is increasing, and a stacked semiconductor device formed by stacking multiple individual semiconductor dies, such as three-dimensional integrated circuit (3DIC), is being developed as such packaging technology.

[0004] Stacked semiconductor devices are manufactured by bonding same or different types of semiconductor chips with connection members. In the case of a stacked semiconductor device in which the pitch of the connection members is designed to be about 21 μm to about 100 μm , bumps and a nonconductive film (NCF) are used to bond the semiconductor dies. In the case of a stacked semiconductor device in which the pitch of the connection members is designed to be about 4 μm to about 20 μm , a hybrid bonding process is performed to bond the semiconductor dies.

[0005] The hybrid bonding process may be applied to a stacking process to manufacture a stacked semiconductor device that cannot use bumps and nonconductive film (NCF) and requires ultra-fine pitch. However, the hybrid bonding process requires high-precision equipment to perform hybrid bonding, and there is an issue in that the bonding surface of each semiconductor die must be managed to have a surface roughness of about 10 $\text{\AA}$ or less to prevent connection defects.

SUMMARY OF THE INVENTION

[0006] In a semiconductor package including a first semiconductor die, a plurality of bonding pads on the first semiconductor die, connection members on the bonding pads, and a second semiconductor die on the connection members, an anisotropic conductive film (ACF) may be disposed between the first semiconductor die and the second semiconductor die.

[0007] The anisotropic conductive film includes conductive particles locally positioned between each of the connection members and a corresponding bonding pad of the bonding pads, and around each of the connection members and a corresponding bonding pad of the bonding pads. The conductive particles may electrically connect each of the connection members to the corresponding bonding pad of the bonding pads.

[0008] A semiconductor package according to an embodiment, includes a first semiconductor die, a plurality of bonding pads on the first semiconductor die, a plurality of connection members on the plurality of bonding pads, wherein each of the plurality of connection members is disposed on a corresponding bonding pad of the plurality of bonding pads, a second semiconductor die on the plurality of connection members, and an anisotropic conductive film between the first semiconductor die and the second semiconductor die, and between the plurality of bonding pads and the plurality of connection members, including an insulating film, and conductive particles inside the insulating film, wherein the conductive particles are positioned between each of the plurality of connection members and the corresponding bonding pad of the plurality of bonding pads.

[0009] A semiconductor package according to an embodiment includes a first semiconductor die, a second semiconductor die on the first semiconductor die, and an interconnection structure between the first semiconductor die and the second semiconductor die, wherein the interconnection structure includes a plurality of bonding pads on the first semiconductor die, a plurality of connection members under the second semiconductor die, wherein each connection member of the plurality of connection members is disposed on a corresponding bonding pad of the plurality of bonding pads, and an anisotropic conductive film surrounding the plurality of bonding pads and the plurality of connection members, wherein the anisotropic conductive film comprises first regions and a second region other than the first regions, wherein the first regions and the second region are defined by dividing a plane of the anisotropic conductive film, each first region of the first regions is defined around the corresponding bonding pad of the plurality of bonding pads, around a corresponding connection member of the plurality of connection members, and between the corresponding bonding pad of the plurality of bonding pads and the corresponding connection member of the plurality of connection members, the anisotropic conductive film includes an insulating film and conductive particles inside the insulating film, and the conductive particles are positioned only in the first regions.

[0010] A method for manufacturing a semiconductor package includes forming an anisotropic conductive film by inserting conductive particles into each first region of first regions of an insulating film, wherein the insulating film comprises first regions and a second region other than the first regions, and the first regions and the second region are defined by dividing a plane of the insulating film, attaching the anisotropic conductive film on an upper semiconductor die on which a plurality of connection members are disposed, wherein each first region of the first regions covers a corresponding connection member of the plurality of connection members, aligning the upper semiconductor die on which the plurality of connection members are disposed on a lower semiconductor die on which a plurality of bonding pads are disposed, and inserting each of the plurality of bonding pads into each first region of the first regions, and bonding each of the plurality of bonding pads to a corresponding connection member of the plurality of connection members, wherein the conductive particles are positioned between each of the plurality of bonding pads and the corresponding connection member of the plurality of connection members.

[0011] The anisotropic conductive film (ACF) may be used as an interconnection structure that connects the first semiconductor die and the second semiconductor die of a stacked semiconductor device requiring ultra-fine pitch, and includes locally positioned conductive particles.

[0012] As a result, high-precision equipment required in the hybrid bonding process may not be required, and the surface roughness of the bonding surface of the semiconductor die, which is managed to prevent connection defects that may occur in the hybrid bonding process, may not need

to be managed.

[0013] The conductive particles in the anisotropic conductive film (ACF) are locally positioned between each of the connection members and a corresponding bonding pad of the bonding pads, and around each of the connection members and a corresponding bonding pad of the bonding pads. Therefore, a reliable electrical connection may be made between the connection members and the bonding pads, even when increasing the content of conductive particles within the anisotropic conductive film (ACF) or minimizing the size of the conductive particles, connection defects that short-circuit the circuit do not occur.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0014] FIG. 1 is a cross-sectional view illustrating a semiconductor stacking structure according to an embodiment.

[0015] FIG. 2 is an enlarged cross-sectional view of region A of the semiconductor stacking structure of FIG. 1.

[0016] FIG. 3 is a cross-sectional view illustrating a semiconductor package according to an embodiment.

[0017] FIG. 4 is a cross-sectional view illustrating a semiconductor package according to another embodiment.

[0018] FIGS. 5 to 12 are cross-sectional views illustrating a method of manufacturing an anisotropic conductive film (ACF) according to an embodiment.

[0019] FIGS. 13 to 16 are cross-sectional views illustrating a method of manufacturing the semiconductor stacking structure according to an embodiment of FIG. 1.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0020] Hereinafter, the present disclosure will be described in detail hereinafter with reference to the accompanying drawings, in which embodiments of the present disclosure are shown. As those skilled in the art would realize, the described embodiments may be modified in various different ways, all without departing from the spirit or scope of the present disclosure.

[0021] The drawings and description are to be regarded as illustrative in nature and not restrictive. Like reference numerals designate like elements throughout the specification.

[0022] Size and thickness of each constituent element in the drawings may be arbitrarily illustrated for better understanding and ease of description, the following embodiments are not limited thereto.

[0023] Throughout the specification, the term “connected” does not mean only that two or more constituent components are directly connected, but may also mean that two or more constituent components are indirectly connected through another constituent component. Throughout the specification, when a component is described as “comprising” “including” (or variations thereof) a particular element or group of elements, it is to be understood that the component may be formed of only the element or the group of elements, or the element or group of elements may be combined with additional elements to form the component, unless the context indicates otherwise. The term “consisting of,” on the other hand, indicates that a component is formed only of the element(s) listed.

[0024] It will be understood that when an element such as a layer, film, region, or substrate is referred to as being “on” another element, it can be directly on the other element or intervening elements may also be present. In contrast, when an element is referred to as being “directly on” another element, the elements are in contact with each other and there are no intervening elements present at the point of contact. Further, when an element is referred to as being “above” or “on” a reference element, it can be positioned above or below the reference element, and it is not necessarily referred to as being positioned “above” or “on” in a direction opposite to gravity.

[0025] Further, throughout the specification, the phrase “in a plan view” refers to viewing a target portion from the top, and the phrase “in a cross-sectional view” or “on a cross-section” means viewing a cross-section formed by vertically cutting a target portion from the side.

[0026] As used herein, components described as being “electrically connected” are configured such that an electrical signal can be transferred from one component to the other (although such electrical signal may be attenuated in strength as it is transferred and may be selectively transferred). Moreover, components that are “directly electrically connected” form a common electrical node through electrical connections by one or more conductors, such as, for example, wires, pads, internal electrical lines, through vias, etc. As such, directly electrically connected components do not include components electrically connected through active elements, such as transistors or diodes.

[0027] Hereinafter, a semiconductor stacking structure **100**, semiconductor packages **10** and **20**, and a method of manufacturing the semiconductor stacking structure **100** according to an embodiment will be described with reference to the drawings.

[0028] FIG. **1** is a cross-sectional view illustrating the semiconductor stacking structure **100** according to an embodiment. FIG. **2** is an enlarged cross-sectional view of region A of the semiconductor stacking structure **100** of FIG. **1**.

[0029] Referring to FIG. **1**, the semiconductor stacking structure **100** includes a first semiconductor die (lower semiconductor die) **110**, an interconnection structure **120**, and a second semiconductor die (upper semiconductor die) **130**. In an embodiment, the semiconductor stacking structure **100** may be a stacking structure of semiconductor dies used in high bandwidth memory (HBM) or 3DIC. In an embodiment, the semiconductor stacking structure **100** may be manufactured based on fan out wafer level package (FOWLP) or fan out panel level package (FOPLP) technology.

[0030] In an embodiment, a first semiconductor die **110** and a second semiconductor die **130** may be the same type (e.g., identical). In an embodiment, the first semiconductor die **110** and the second semiconductor die **130** may be different type.

[0031] The first semiconductor die **110** includes a first die base **111**, a first dielectric layer **112**, and through substrate vias **113**. The first die base **111** includes a substrate having active regions formed therein, and a wiring layer on the active regions. The first die base **111** has a front side **111F** corresponding to the side of the substrate in which the active regions are formed (and corresponding to the side on which the wiring layer is formed). The front side of the die base **111** corresponds to the front side of the first semiconductor die **110**. The first die base **111** has a back side **111B** opposite to the front side **111F**, the back side **111B** corresponding to the back side of the first semiconductor die **110**). The wiring layer may include several metal layers separated by interlayer dielectric layers, where patterns of the metal layers are interconnected to form wiring. The active regions may be used to form transistors that are interconnected by wiring of the wiring layer to form logic circuits (e.g., NAND, NOR, inverters, etc.). The logic circuits may be interconnected by the wiring of the wiring layer to form an integrated circuit of the first semiconductor die **110**, such as a memory integrated circuit or a logic device integrated circuit. The wiring layer includes signal wiring that transmits signals between the first semiconductor die **110** and external devices (such as the second semiconductor die **130**) and power wiring that transmits power to the first semiconductor die **110**. The backside **111B** of the first die base **111** may be the backside of the first semiconductor die **110** and may not have any active regions formed therein, nor any transistors formed with the backside **111B** of the substrate of the first semiconductor die **110**.

[0032] The term “substrate” may denote a base substrate (e.g., an initial semiconductor substrate forming the base of the wafer in the final wafer product, such as a bulk semiconductor substrate (e.g., formed of crystalline silicon), a silicon on insulator (SOI) substrate, etc.).

[0033] The first dielectric layer **112** is formed on the back side of the substrate. In an embodiment, the first dielectric layer **112** may include at least one of a silicon oxide layer and a silicon nitride

layer. The first dielectric layer may be a passivation layer and forms an external surface of the semiconductor die **110**.

[0034] The through substrate vias (TSV) **113** (which may be through silicon vias) penetrate the first dielectric layer **112** and contact the bonding pads **123**. The through substrate vias (TSV) **113** extend through the substrate and wiring layer of the first die base **111** and may extend fully through the first semiconductor chip **110** to the bonding pads **123**. Each of the through substrate vias (TSV) **113** is electrically connected to a corresponding bonding pad of the bonding pads **123**. In an embodiment, the through substrate vias (TSV) **113** may include at least one of tungsten, aluminum, copper, and alloys thereof. In an embodiment, an insulating layer (not shown) may be formed around a side surface of each of the through substrate vias (TSV) **113**. In an embodiment, an insulating layer (not shown) may be formed between the through substrate vias (TSV) **113** and the substrate, and between the through substrate vias (TSV) **113** and the first dielectric layer **112**. In an embodiment, the insulating layer (not shown) may include at least one of an oxide layer, a nitride layer, a carbonization layer, and a polymer. In an embodiment, the insulating layer (not shown) may be an ozone/tetra-ethyl ortho-silicate (O₃/TEOS)-based high aspect ratio process (HARP) oxide layer formed by a sub-atmospheric chemical vapor deposition (CVD) process.

[0035] The interconnection structure **120** is disposed between the first semiconductor die **110** and the second semiconductor die **130**. The interconnection structure **120** is disposed between the back side **111B** of the first semiconductor die **110** and the front side **131F** of the second semiconductor die **130**. The interconnection structure **120** includes an anisotropic conductive film (ACF) **121A**, bonding pads **123**, and connection members **127**.

[0036] The anisotropic conductive film (ACF) **121A** is disposed between the first semiconductor die **110** and the second semiconductor die **130**, and between the bonding pads **123** and the connection members **127**. The anisotropic conductive film (ACF) **121A** surrounds the bonding pads **123** and the connection members **127**. The anisotropic conductive film (ACF) **121A** includes an insulating film **121** and conductive particles **122**. The insulating film **121** is adhered to the first semiconductor die **110** and the second semiconductor die **130** to fix the first semiconductor die **110** to the second semiconductor die **130**, and has the function of insulating and protecting the bonding pads **123** and the connection members **127**. In an embodiment, the insulating film **121** may include a polymer film. In an embodiment, the insulating film **121** may include thermosetting resin. In an embodiment, the insulating film **121** may include acrylic resin or epoxy resin. In an embodiment, the insulating film **121** may be a non-conducting film (NCF).

[0037] The anisotropic conductive film (ACF) **121A** includes first regions **R1** and a second region **R2**. The second region **R2** may be the portion of the anisotropic conductive film (ACF) **121A** that is not the first regions **R1**. Although not shown in the cross sectional views of the figures, the second region **R2** may be continuous while the first regions **R1** may be separate from each other, being separated by the second region **R2**. With respect to a top down view, the first regions may be islands within the second region **R2**. Each first region **R1** may be a region surrounding a corresponding one of the bonding pads **123**, a corresponding one of the connection members **127**, and the region therebetween. In an embodiment, with respect to a plan view, each of the first regions **R1** may have a circular, oval, polygonal, or conformal shape around each of the connection members **127**.

[0038] The conductive particles **122** are included inside the insulating film **121**. Referring to FIG. 2, the conductive particles **122** are positioned within each first region of the first regions **R1**. The conductive particles **122** are not positioned in the second region **R2**. The conductive particles **122** are positioned between each connection member **127** and the bonding pad **123** adjacent to this connection member **127**. The conductive particles **122** may electrically connect each of the connection members **127** to a corresponding bonding pad among the bonding pads **123**. With respect to plan view, the conductive particles **122** are positioned to surround each of the bonding pads **123** and to surround each of the connection members **127**.

[0039] Some of the conductive particles **122** contact the side surface of a bonding pad **123**. Some of the conductive particles **122** contact the side surface of a connection member **127**.

[0040] The conductive particles **122** are positioned within a predetermined distance in the horizontal direction from the side surface of a corresponding bonding pad **123**. The conductive particles **122** are positioned within a predetermined distance in the horizontal direction from the side surface of a corresponding connection member **127**.

[0041] In an embodiment, each of the conductive particles **122** may be metal particles coated with a conductive metal layer, or polymer particles coated with a conductive metal layer. In an embodiment, the metal particles may be formed of a metal material with high hardness. In an embodiment, the metal particle may include nickel, solder, or copper. In an embodiment, the polymer particles may be polymer balls. In an embodiment, the conductive metal layer may be formed by performing an electrolytic plating process, an electroless plating process, or a chemical vapor deposition (CVD) process on the metal particles or the polymer particles. In an embodiment, the conductive metal layer may include at least one of nickel, gold, silver, copper, platinum, palladium, cobalt, tin, indium, and indium tin oxide (ITO). In an embodiment, each of the conductive particles **122** may have a spherical or non-spherical shape. In an embodiment, the conductive particles **122** may have a diameter D in the range of 10 nanometers (nm) to 20 micrometers (μm).

[0042] The bonding pads **123** are disposed on the first dielectric layer **112** of the first semiconductor die **110**. Each of the bonding pads **123** is electrically connected to a corresponding through silicon via (TSV) **113**. In an embodiment, the horizontal pitch P between neighboring bonding pads **123** may be in the range of 4 μm to 100 μm . Referring to FIG. 2, the bonding pad **123** includes a first pad layer **124**, a second pad layer **125**, and a third pad layer **126**.

[0043] The first pad layer **124** is disposed on and contacts the through silicon via (TSV) **113** and is electrically connected to the through silicon via (TSV) **113**. The first pad layer **124** includes a barrier layer and a seed metal layer. The barrier layer is made of a material that is difficult to diffuse into the first dielectric layer **112** and prevents the metal component of the seed metal layer from diffusing into the first dielectric layer **112**. In an embodiment, the barrier layer may include titanium, tantalum, a titanium alloy, a tantalum alloy, a titanium nitride layer, or a tantalum nitride layer. In an embodiment, the barrier layer may be formed by performing a sputtering process. In an embodiment, the seed metal layer may include copper. In an embodiment, the seed metal layer may be formed by performing an electroless plating process. In an embodiment, a cleaning process or a metal catalyst activation pretreatment process may be performed prior to electroless plating. In an embodiment, the seed metal layer may be formed by performing a sputtering process.

[0044] A second pad layer (nickel layer) **125** is disposed on the first pad layer **124**. The second pad layer **125** serves to reduce the resistance of the bonding pad **123**. In an embodiment, the second pad layer **125** may include nickel. In an embodiment, the second pad layer **125** may be formed by performing an electroplating process. In an embodiment, the second pad layer **125** may have a vertical dimension, e.g., a thickness, greater than the vertical dimension, e.g., the thickness, of the first pad layer **124** and the vertical dimension, e.g., the thickness, of the third pad layer **126**.

[0045] A third pad layer (gold layer) **126** is disposed on the second pad layer **125**. The third pad layer **126** serves to improve adhesion to the conductive particles **122**. In an embodiment, the third pad layer **126** may include gold. In an embodiment, the third pad layer **126** may be formed by performing an electroplating process. In an embodiment, the third pad layer **126** may have a vertical dimension, e.g., a thickness, that is thinner than the vertical dimension, e.g., the thickness, of the second pad layer **125**.

[0046] According to the present disclosure, the bonding pad **123** including the first pad layer **124**, the second pad layer **125**, and the third pad layer **126** is described, but is not limited thereto, and the bonding pad **123** including fewer or more pad layers may be included within the present disclosure as needed.

[0047] The connection members **127** are disposed below the second dielectric layer **134** of the second semiconductor die **130**. Each of the connection members **127** is disposed on a corresponding one of the bonding pads **123**. Each of the connection members **127** electrically connects a corresponding one of the contact vias **135** of the second semiconductor die **130** to the corresponding one of bonding pads **123** of the first semiconductor die **110** via the conductive particles **122**.

[0048] In an embodiment, the horizontal pitch P between neighboring connection members among the connection members **127** may be 4 μm to about μm . In an embodiment, the connection members **127** may include micro bumps (e.g., conductive balls or pillars). In an embodiment, the connection members **127** may include at least one of copper, aluminum, tungsten, nickel, gold, silver, tin, titanium, and alloys thereof.

[0049] The second semiconductor die **130** includes a second die base **131**, a second dielectric layer **134**, and contact vias **135**. The second die base **131** includes a substrate, active regions formed in the substrate, and a wiring layer on the active regions. The second die base **131** has the front side **131F** corresponding to the side of the substrate in which the active regions are formed (and corresponding to the side on which the wiring layer is formed). The front side of the second die base **131** corresponds to the front die of the second semiconductor die **130**. The second die base has back side **131B** opposite to the front side **131F**, the back side **131** corresponding to the back side of the second semiconductor die **130**. The wiring layer and active regions of the second semiconductor die may form transistors, logic circuits and an integrated circuit (e.g. an integrated circuit of a memory device or an integrated circuit of a logic device) as described with respect to the first semiconductor die **110** (i.e., with respect to the front side of the second semiconductor die **130**). Similarly, the backside of the second semiconductor die **130** may not be provided with transistors. The wiring layer includes signal wiring that transmits signals between the second semiconductor die **130** and external devices and power wiring that transmits power to the second semiconductor die **130**.

[0050] The second dielectric layer **134** is positioned on the front side **131F** of the substrate and is formed to cover the wiring layer. In an embodiment, the second dielectric layer **134** may include a silicon nitride layer. The contact vias **135** penetrate the second dielectric layer **134** and contact the connection members **127**. Each of the contact vias **135** is disposed between a bonding pad (not shown) and each of the connection members **127**. Each of the contact vias **135** is electrically connected to each of the connection members **127**. In an embodiment, the contact vias **135** may include at least one of copper, aluminum, tungsten, nickel, gold, silver, tin, titanium, and alloys thereof.

[0051] According to the present disclosure, the conductive particles **122** may be disposed only in first regions R1 of the anisotropic conductive film (ACF) **121A** which are regions surrounding the bonding pads **123**, surrounding each of the connection members **127**, and between each of the bonding pads **123** and each of the connection members **127**.

[0052] By disposing the conductive particles **122** only in the first regions R1 as described above, the content of the conductive particles **122** per unit area of the insulating film **121** within the first regions R1 increases significantly, thereby improving electrical characteristics between the bonding pads **123** and the connection members **127**. In addition, the size of the conductive particles **122** in the first regions R1 may be reduced to significantly increase the contact area between the conductive particles **122**, thereby improving the electrical characteristics between the bonding pads **123** and the connection members **127**. The content of conductive particles **122** in the second region R2 is 0 or substantially 0.

[0053] Contrary to the present disclosure, if the content of conductive particles **122** within the insulating film **121** is increased, or the size of the conductive particles **122** is decreased, to improve electrical characteristics, without limiting the regions where the conductive particles **122** are disposed, the conductive particles **122** may become spread out within the insulating film **121**,

resulting in poor contact between the bonding pads **123** and the connection members **127**.

[0054] In this way, by disposing the conductive particles **122** only in the first regions **R1** of the anisotropic conductive film (ACF) **121A** for the electrical connection between the bonding pads **123** and the connection members **127**, it is possible to apply the bonding process of the anisotropic conductive film (ACF) **121A** according to the present disclosure, without applying a hybrid bonding process to the semiconductor stacking structure **100** where the pitch of the connection members is designed to be ultra-fine pitch (for example, in a range of 4 μm to 20 μm). Accordingly, high-precision equipment required in the hybrid bonding process may not be required, and the surface roughness of the bonding surface of the semiconductor die, which is managed to prevent connection defects that may occur in the hybrid bonding process, does not have to be managed.

[0055] FIG. **3** is a cross-sectional view illustrating a semiconductor package **10** according to an embodiment.

[0056] Referring to FIG. **3**, the semiconductor package **10** includes a redistribution structure **510**, the semiconductor stacking structure **100** on the redistribution structure **510**, conductive posts **600** on the redistribution structure **510**, and a molding material **610** that covers the first semiconductor die **110** on the redistribution structure **510**. The semiconductor stacking structure **100** may be that described with respect to FIGS. **1** and **2** and includes the first semiconductor die **110**, the interconnection structure **120**, and the second semiconductor die **130**. In an embodiment, the semiconductor package **10** may include a system on chip (SoC). In an embodiment, the semiconductor package **10** may include a three-dimensional integrated circuit (3DIC). In an embodiment, the first semiconductor die **110** may be the lower die of a 3D integrated circuit. In an embodiment, the second semiconductor die **130** may be the upper die of a 3D integrated circuit. The footprint of the first semiconductor die **110** is included within the footprint of the second semiconductor die **130**. The second semiconductor die **130** has a larger horizontal area (i.e., with respect to a plan view) than the first semiconductor die **110**.

[0057] The first semiconductor die **110** is a different die from a second semiconductor die **130**. The semiconductor package **10** has a structure in which different types of dies are stacked. In an embodiment, the first semiconductor die **110** may include a central processing unit (CPU) or a graphic processing unit (GPU). In an embodiment, the second semiconductor die **130** may include at least one of a memory, a communication chip, a controller, and a sensor.

[0058] In the semiconductor package **10** manufactured based on fan out wafer level package (FOWLP) or fan out panel level package (FOPLP) technology, the interconnection structure **120** connects the second semiconductor die **130** to the first semiconductor die **110**, and connects the second semiconductor die **130** to the conductive posts **600**. In this case, the pitch between neighboring bonding pads among the bonding pads **123** connected to the first semiconductor die **110** and the pitch between neighboring connection members among the connection members **127** connected thereto are smaller than the pitch between neighboring bonding pads among the bonding pads **123** connected to the conductive posts **600** and the pitch between neighboring connection members among the connection members **127** connected thereto.

[0059] FIG. **4** is a cross-sectional view illustrating a semiconductor package **20** according to another embodiment.

[0060] Referring to FIG. **4**, the semiconductor package **20** includes a plurality of semiconductor stacking structures **100**. The semiconductor package **20** has a structure in which first semiconductor dies **110** are alternately stacked. Each adjacent pair of first semiconductor dies **110** and the interconnection structure **120** therebetween may correspond to the semiconductor stacking structures **100** described herein with respect to FIGS. **1** and **2**. The semiconductor package **20** has a structure in which dies of the same type are stacked. In an embodiment, the semiconductor package **20** may be high bandwidth memory (HBM).

[0061] The high bandwidth memory (HBM) is a high-performance three-dimensional (3D) stacked dynamic random-access memory RAM (DRAM). The high bandwidth memory (HBM) is

manufactured by stacking memory dies vertically to form a single memory stack. The high bandwidth memory (HBM) may implement shorter latency and higher bandwidth compared to conventional DRAM products by stacking memory dies vertically to have multiple memory channels, while reducing the total area occupied by individual DRAMs on a printed circuit board (PCB), which is advantageous for high bandwidth to area and may reduce power consumption.

[0062] The high bandwidth memory (HBM) is designed so that the connection members connecting the memory dies have ultra-fine pitches. According to the present disclosure, the anisotropic conductive film (ACF) **121A** on which the conductive particles **122** are locally disposed is used, thereby connecting memory dies of the high bandwidth memory (HBM).

[0063] The semiconductor package **20** includes a buffer die **700**, the first semiconductor dies **110**, the interconnection structure **120**, and a molding material **800**. The buffer die **700** is disposed at the bottom of the semiconductor package **20**. When data is exchanged between devices with different data processing speeds, processing units, and usage times, data loss may occur due to differences in data processing speeds, processing units, and usage times between each device. To prevent such loss, the buffer die **700** is included, and when exchanging data between the first semiconductor dies **110** and an external device, such data is temporarily stored in the buffer die **700** (e.g., in buffers of the buffer die **700**). When transmitting data to or receiving data from the first semiconductor dies **110**, the buffer die **700** arranges the order of the data and then sequentially passes the data to its appropriate destination.

[0064] The first semiconductor dies **110** are disposed on the buffer die **700**. The first semiconductor die **110** on which no through substrate vias are formed is disposed at the top of the memory stack. The interconnection structure **120** is disposed between the first semiconductor die **110** and the buffer die **700** and between the first semiconductor dies **110**. The first semiconductor die **110** and the buffer die **700**, and the neighboring first semiconductor dies **110** are bonded to each other by the anisotropic conductive film (ACF) **121A** on which the conductive particles **122** are locally disposed.

[0065] The molding material **800** is disposed on the buffer die **700** and covers the first semiconductor dies **110** and the interconnection structures **120**. The molding material **800** serves to protect and insulate the first semiconductor dies **110** and the interconnection structures **120**. In an embodiment, the molding material **800** may be formed of a thermosetting resin such as epoxy resin. In an embodiment, the molding material **800** may be an epoxy molding compound (EMC).

[0066] FIGS. **5** to **12** are cross-sectional views illustrating a method of manufacturing the anisotropic conductive film (ACF) **121A** according to an embodiment.

[0067] FIG. **5** is a cross-sectional view illustrating providing the insulating film **121** to manufacture the anisotropic conductive film (ACF) **121A**.

[0068] Referring to FIG. **5**, the insulating film **121** is attached on a substrate **129**. A separation layer **128** is positioned on the substrate **129**. The separation layer **128** is a surface-treated layer of the upper surface of the substrate **129**. In an embodiment, the substrate **129** may include a slide glass. In an embodiment, the separation layer **128** may be a layer surface-treated with silane.

[0069] FIG. **6** is a cross-sectional view illustrating aligning a pillar stamp **900** over the insulating film **121**.

[0070] Referring to FIG. **6**, the pillar stamp **900** is positioned over the insulating film **121** attached to the substrate **129**. The pillar stamp **900** includes convex portions. In an embodiment, the convex portions may have a circular, oval, polygonal, or conformal shape associated with positions of each of the connection members **127**. In an embodiment, the pillar stamp **900** may be a photoresist pattern formed by exposing and developing photoresist and hardening the photoresist by heat treatment. In an embodiment, the pillar stamp **900** may be a polymer pattern obtained by placing and curing a free polymer solution in a mold. In an embodiment, the surface of the pillar stamp **900** on which the convex portions are formed may be surface-treated with silane.

[0071] FIG. **7** is a cross-sectional view illustrating forming recess portions in the insulating film

121 using the pillar stamp **900**.

[0072] Referring to FIG. 7, the insulating film **121** is patterned by the pillar stamp **900**. The pattern of the insulating film **121** is formed by contacting the convex portions of the pillar stamp **900** with the insulating film **121** and thermally compressing the insulating film **121** with the pillar stamp **900** (e.g., moving the pillar stamp in the direction of the insulating film **121** to press the insulating film **121** and heating the insulating film with the pillar stamp). The insulating film **121** is patterned and includes recess portions that match the shape of the convex portions of the pillar stamp **900**. As recess portions are formed in the insulating film **121**, the thickness of the region around the recess portions of the insulating film **121** increases.

[0073] FIG. 8 is a cross-sectional view illustrating removing the pillar stamp **900** from the insulating film **121**.

[0074] Referring to FIG. 8, the pillar stamp **900** is removed from the insulating film **121**. The surface of the pillar stamp **900** on which the convex portions are formed may have been previously surface-treated with silane so that the pillar stamp **900** may be easily removed from the insulating film **121**.

[0075] FIG. 9 is a cross-sectional view illustrating disposing the conductive particles **122** in recess portions of the insulating film **121**.

[0076] Referring to FIG. 9, the conductive particles **122** are attached to an elastic member **950**. The elastic member **950** to which the conductive particles **122** are attached moves on the insulating film **121** and sequentially stops on the recess portions of the insulating film **121**. The conductive particles **122** move from the elastic member **950** into the recess portions of the insulating film **121**. In an embodiment, the elastic member **950** may include polydimethylsiloxane (PDMS).

[0077] FIG. 10 is a cross-sectional view illustrating aligning the substrate **129** on the insulating film **121** on which the conductive particles **122** are disposed.

[0078] Referring to FIG. 10, the substrate **129** is aligned on the insulating film **121** on which the conductive particles **122** are disposed. The separation layer **128** is disposed on the substrate **129**. The separation layer **128** is a surface-treated layer of the upper surface of the substrate **129**. In an embodiment, the substrate **129** may include a slide glass. In an embodiment, the separation layer **128** may be a layer surface-treated with silane.

[0079] FIG. 11 is a cross-sectional view illustrating compressing the insulating film **121** on which the conductive particles **122** are disposed.

[0080] Referring to FIG. 11, the insulating film **121** is thermally compressed using the substrates **129** with the insulating film **121** on which the conductive particles **122** are disposed therebetween. For example, the substrate **129** are used to compress and heat the insulating film. By thermally compressing, the conductive particles **122** are locally inserted into the insulating film **121**, and the anisotropic conductive film (ACF) **121A** is completed.

[0081] FIG. 12 is a cross-sectional view illustrating removing the substrates **129** from the anisotropic conductive film (ACF) **121A**.

[0082] Referring to FIG. 12, the substrates **129** attached to the upper portion and lower portion of the anisotropic conductive film (ACF) **121A** are removed from the anisotropic conductive film (ACF) **121A**. The substrates **129** each include the separation layer **128** so that the substrates **129** may be easily removed from the anisotropic conductive film (ACF) **121A**.

[0083] FIGS. 13 to 16 are cross-sectional views illustrating a method of manufacturing the semiconductor stacking structure **100** according to an embodiment of FIG. 1. The method of manufacturing the semiconductor stacking structure **100** may initially comprise the method of manufacturing the anisotropic conductive film (ACF) **121A** as described with respect to FIGS. 5 to 12 and need not be repeated here.

[0084] FIG. 13 is a cross-sectional view illustrating aligning the anisotropic conductive film (ACF) **121A** on the front side **131F** of the second semiconductor die **130**.

[0085] Referring to FIG. 13, the anisotropic conductive film (ACF) **121A** is aligned on the front

side **131F** of the second semiconductor die **130** so that the conductive particles **122** in the insulating film **121** are positioned on each of the connection members **127**.

[0086] FIG. **14** is a cross-sectional view illustrating attaching the anisotropic conductive film (ACF) **121A** on the front side **131F** of the second semiconductor die **130**.

[0087] Referring to FIG. **14**, the anisotropic conductive film (ACF) **121A** is attached on the front side **131F** of the second semiconductor die **130**. The connection members **127** are inserted into the insulating film **121**. The exposed surfaces of the connection members **127** are surrounded by conductive particles **122**.

[0088] FIG. **15** is a cross-sectional view illustrating aligning the second semiconductor die **130** to which the anisotropic conductive film (ACF) **121A** is attached on the back side **111B** of the first semiconductor die **110**.

[0089] Referring to FIG. **15**, the second semiconductor die **130** with the anisotropic conductive film (ACF) **121A** attached thereto is aligned on the first semiconductor die **110** such that each of the connection members **127** and the conductive particles **122** surrounding each of the connection members **127** are positioned on a corresponding bonding pad of the bonding pads **123**.

[0090] FIG. **16** is a cross-sectional view illustrating the second semiconductor die **130**, the anisotropic conductive film (ACF) **121A**, and the first semiconductor die **110** attached together, which depicts the semiconductor stacking structure **100**.

[0091] Referring to FIG. **16**, the second semiconductor die **130** is attached to the anisotropic conductive film (ACF) **121A**, which is further attached to the first semiconductor die **110**. The bonding pads **123** are inserted into the insulating film **121**. The surface of each bonding pad among the bonding pads **123** is surrounded by conductive particles **122**. Each of the bonding pads **123** is electrically connected to each of the connection members **127** by the conductive particles **122**.

[0092] While this invention has been described in connection with what is presently considered to be practical embodiments, it is to be understood that the invention is not limited to the disclosed embodiments, but, on the contrary, is intended to cover various modifications and equivalent arrangements included within the spirit and scope of the invention.

Claims

1. A semiconductor package, comprising: a first semiconductor die; a plurality of bonding pads on the first semiconductor die; a plurality of connection members on the plurality of bonding pads, the plurality of connection members being conductive, wherein each connection member of the plurality of connection members is disposed on a corresponding bonding pad of the plurality of bonding pads; a second semiconductor die on the plurality of connection members; and an anisotropic conductive film between the first semiconductor die and the second semiconductor die, and between the plurality of bonding pads and the plurality of connection members, the anisotropic conductive film comprising: an insulating film; and conductive particles inside the insulating film, wherein the conductive particles are positioned between each of the plurality of connection members and the corresponding bonding pad of the plurality of bonding pads.
2. The semiconductor package of claim 1, wherein the insulating film comprises a non-conducting film (NCF).
3. The semiconductor package of claim 1, wherein the insulating film comprises a thermosetting resin.
4. The semiconductor package of claim 1, wherein the insulating film comprises acryl resin or epoxy resin.
5. The semiconductor package of claim 1, wherein the conductive particles electrically connect each of the plurality of connection members to a corresponding bonding pad of the plurality of bonding pads.
6. The semiconductor package of claim 1, wherein each of the conductive particles comprises metal

particles coated with a conductive metal layer or polymer particles coated with a conductive metal layer.

7. The semiconductor package of claim 6, wherein the metal particles comprise nickel, solder, or copper.

8. The semiconductor package of claim 6, wherein the conductive metal layer comprises at least one of nickel, gold, silver, copper, platinum, palladium, cobalt, tin, indium, and indium tin oxide (ITO).

9. The semiconductor package of claim 1, wherein the plurality of connection members comprise micro bumps.

10. The semiconductor package of claim 1, wherein each of the plurality of bonding pads comprises: a seed metal layer; a nickel layer on the seed metal layer; and a gold layer on the nickel layer.

11. The semiconductor package of claim 1, wherein a horizontal pitch between the connection members is in the range of 4 micrometers to 100 micrometers.

12. A semiconductor package, comprising: a first semiconductor die; a second semiconductor die on the first semiconductor die; and an interconnection structure between the first semiconductor die and the second semiconductor die, wherein the interconnection structure comprises: a plurality of bonding pads on the first semiconductor die; a plurality of connection members under the second semiconductor die, the plurality of connection members being conductive, wherein each connection member of the plurality of connection members is disposed on a corresponding bonding pad of the plurality of bonding pads; and an anisotropic conductive film surrounding the plurality of bonding pads and the plurality of connection members, wherein the anisotropic conductive film comprises first regions and a second region other than the first regions, each first region of the first regions surrounds a corresponding bonding pad of the plurality of bonding pads with respect to a top down view, surrounds a corresponding connection member of the plurality of connection members with respect to the top down view, and is positioned between the corresponding bonding pad of the plurality of bonding pads and the corresponding connection member of the plurality of connection members, the anisotropic conductive film comprises an insulating film and conductive particles inside the insulating film, and the conductive particles are positioned in the first regions and not in the second region.

13. The semiconductor package of claim 12, wherein within each of the first regions, some of the conductive particles contact side surfaces of the corresponding bonding pad among the plurality of bonding pads, and some of the conductive particles contact the side surfaces of the corresponding connection member among the plurality of connection members.

14. The semiconductor package of claim 12, wherein the first semiconductor die is the same type as the second semiconductor die.

15. The semiconductor package of claim 14, wherein the first semiconductor die and the second semiconductor die are dynamic random access memory (DRAM).

16. The semiconductor package of claim 12, wherein the first semiconductor die forms a different type of integrated circuit from that of the second semiconductor die.

17. The semiconductor package of claim 16, wherein the first semiconductor die comprises a central processing unit (CPU) or a graphic processing unit (GPU).

18. The semiconductor package of claim 16, wherein the second semiconductor die comprises at least one of a memory, a communication chip, a controller, and a sensor.

19. A method for manufacturing a semiconductor package, comprising: forming an anisotropic conductive film by inserting conductive particles into each first region of first regions of an insulating film, wherein the insulating film comprises the first regions and a second region other than the first regions; attaching the anisotropic conductive film on an upper semiconductor die on which a plurality of connection members are disposed, the plurality of connection members being conductive, wherein each first region of the first regions aligns with a corresponding connection

member of the plurality of connection members; aligning the upper semiconductor die on which the plurality of connection members are disposed with a lower semiconductor die on which a plurality of bonding pads are disposed; and inserting each of the plurality of bonding pads into a corresponding first region of the first regions, and bonding each of the plurality of bonding pads to a corresponding connection member of the plurality of connection members, wherein at least some of the conductive particles are positioned between each of the plurality of bonding pads and the corresponding connection member of the plurality of connection members.

20. The method for manufacturing the semiconductor package of claim 19, wherein the inserting of conductive particles into each first region of the first regions of the insulating film comprises: forming a plurality of recess portions in the insulating film by pressing a filler stamp on the insulating film; and disposing the conductive particles within each of the plurality of recess portions.
